

RESEARCH ARTICLE

Real-Time Posture Correction of Squat Exercise: A Deep Learning Approach for Performance Analysis and Error Correction

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ABSTRACT The squat is a widely performed exercise essential for maintaining physical fitness and overall well-being. It involves a series of movements in which an individual lowers their torso by bending the knees, and then returns to the starting position. Incorrect squat execution can lead to physical injuries, highlighting the importance of expert guidance. However, many individuals cannot afford professional training due to high costs, necessitating an automated workout analysis system. Existing vision-based squat analysis models face challenges related to occlusion and environmental factors. To address these limitations, we propose a method that utilizes squat videos captured by a stereo camera. The MediaPipe algorithm extracts and tracks keypoints from video frames. Detected keypoints are then used to classify squat types and evaluate squat performance using a Bi-CGRU model. Additionally, we introduce a regression-based approach to predict the correct squat posture. A deep learning-based regressor, trained on a dataset of squat videos, estimates the correct joint angles. When an improper squat is detected, the system provides feedback by presenting the correct squat form to the participant. Experimental results demonstrate that the Bi-CGRU classification model achieves 96.1% accuracy, while the pose correction regression model attains 99.0% accuracy using mean squared error loss. Both qualitative and quantitative evaluations indicate that our approach outperforms state-of-the-art methods utilizing stereo video input. The system helps users to refine their squat technique and reduce the risk of injury by offering real-time posture correction based on biomechanics guidance.

INDEX TERMS Convolutional gated recurrent unit (Bi-CGRU), regression, MediaPipe, direct linear transform (DLT).

I. INTRODUCTION

Performing exercises incorrectly can lead to injury, emphasizing the importance of supervision by a professional. Squats involve lowering the body while maintaining an upright torso and bending the knees, followed by returning to a standing position. However, squat varies among individuals due to differences in femur, torso and limb proportions. Research has shown that squat mechanics are influenced by muscle strength and joint flexibility. For instance, individuals with

a short torso and long femurs perform squats differently from those with a long torso and short femurs. These variations, however, do not indicate improper form but rather natural biomechanical differences. Similarly, individuals with wider hips may squat lower with a wider stance, whereas those with average proportions may find this challenging. However, the joint angles are expected to lie within permissible limits in any of these cases.

Real-time feedback enables people to improve their movement patterns, guaranteeing that they perform exercises correctly by targeting muscle areas. This optimizes the effect of training and speeds up fitness objectives. Immediate

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feedback improves motor learning and skill acquisition by reinforcing proper movement patterns. This is especially helpful for people new to a particular exercise training. Exercise can become more dynamic and interesting with real-time feedback. The progress in correcting the exercise form and technique can increase motivation and training commitment. Personalized feedback and exercise parameter modifications based on each person's unique biomechanics, fitness level, and objectives are made possible by real-time data. This customized strategy lowers the chance of injury while optimizing training results.

Before an injury occurs, real-time systems can identify incorrect movement patterns. The feedback reduces stress on muscles and joints by guaranteeing correct form [1]. This is essential for avoiding injuries, particularly while performing repetitive or high-impact workouts. Normally, real-time information on posture, muscle activity, and body movement is provided via accelerometers, inertial measurement units, and other wearable sensors. Computer vision systems analyze video to evaluate exercise forms and offer feedback. Artificial intelligence systems analyze sensor data and camera data to detect improper movement patterns and offer tailored feedback. There are several ways to provide real-time input, such as haptic feedback, visual displays, and audio signals. Fitness training and injury prevention could be completely transformed by real-time exercise correction. It enables people to exercise safely and efficiently by giving them quick feedback and tailored advice.

Squat has many advantages, including stronger knees and hips and better leg muscles. Further, it strengthens bones, lowering the chance of osteoporosis and the risk of heart-related diseases. Models that use sensors to guide exercise are discussed in the literature [2]. Each joint has sensors, including accelerometers or gyroscopes, that may record joint motion and determine how well an activity is being performed. Additionally, extensive research has been conducted on the use of RGB cameras and depth cameras, such as Kinect, to estimate the user's joint positions accurately. To evaluate squat performance, the combination of camera-based systems and machine learning algorithms has also been studied. The AI fitness trainer [3] employs MediaPipe for pose estimation to assess the squats. This classifies squats as good or bad based on side-view video and using measures like the knee angle between the ankle and hip. Additionally, there are beginners and advanced settings that alter the threshold for squat categorisation. The authors [3] and [4] employ MediaPipe and a hardcoded classification method based on hip and knee positions using side-view videos. The work in [5] uses a pose estimation method for squat classification that is enhanced by a Convolutional Neural Network (CNN) [6]. However, it accepts videos taken from the side for classification. The work [7] uses a custom Neural Networks (NN) regressor in conjunction with the pose estimation technique [8] to classify squats into quarter squats, half squats, and full squats based on depth.

This technique is limited to side-view videos and cannot be categorised in any other way. Squats are divided into six categories with real-time classification and user feedback in the research [9], which uses MediaPipe for pose estimation in conjunction with a Support Vector Machine (SVM) classifier and supports both side and front view footage. The study [10] supports both front and side-view videos for classification and use MediaPipe for pose extraction with a Long Short-Term Memory (LSTM) model to categorize squats into seven categories [11].

Limb length and joint angle data are extracted using pose estimation techniques. Various models, including You Only Look Once v7 (YOLOv7), MediaPipe Pose, OpenPose, VoxelPose, and BlaisePose, have been developed to estimate human poses from images or videos by providing body joint coordinates. Machine learning algorithms such as Hidden Markov Model (HMM), NN, and SVM have been used to analyze exercise performance by leveraging pose estimation-based feature extraction. The model presented in [12], based on a Deep Neural Network (DNN), demonstrates superior performance compared to traditional Convolutional Neural Network (CNN)s for human pose recognition. The methodology in [13] investigates machine learning techniques for human identification and interaction, while [14] presents a 3 Dimensional (3D) pose estimation approach. Multicamera video-based position analysis and human pose estimation are explored in [15]. To enhance pose stability, [16] introduces a technique for estimating the 3D human pose from monocular images by incorporating joint angles. In [17], a method is used to align a human mesh with an actual subject image. Furthermore, [6] examines the use of part affinity fields for multi-person pose estimation, whereas [8] presents an High Resolution Net (HRNet) based approach that maintains high resolution representation during posture estimation.

In [18], the BlazePose technique is utilized to estimate the pose in an exercise evaluation system. To minimize exercise-related injuries, the authors in [19] propose integrating Artificial Intelligence (AI) into fitness assistants that utilize MediaPipe for pose estimation. The concept of a fitness companion designed to allow users to exercise safely is explored in [20]. Using the MediaPipe pose estimation method, [21] presents an AI-based trainer capable of identifying posture and providing corrective feedback. Furthermore, [22] employs the Bayesian estimation method and the Microsoft Kinect camera for body posture analysis. The framework described in [23] estimates the human pose by training a CNN model on Common Object in Content (COCO) datasets. In [24], a mobile fitness assistant is developed utilizing a deep neural network, while [25] applies deep neural networks for real-time fitness training and posture correction.

Squat analysis commonly employs machine learning techniques to classify squat postures based on data extracted from pose estimation algorithms. The use of MediaPipe and

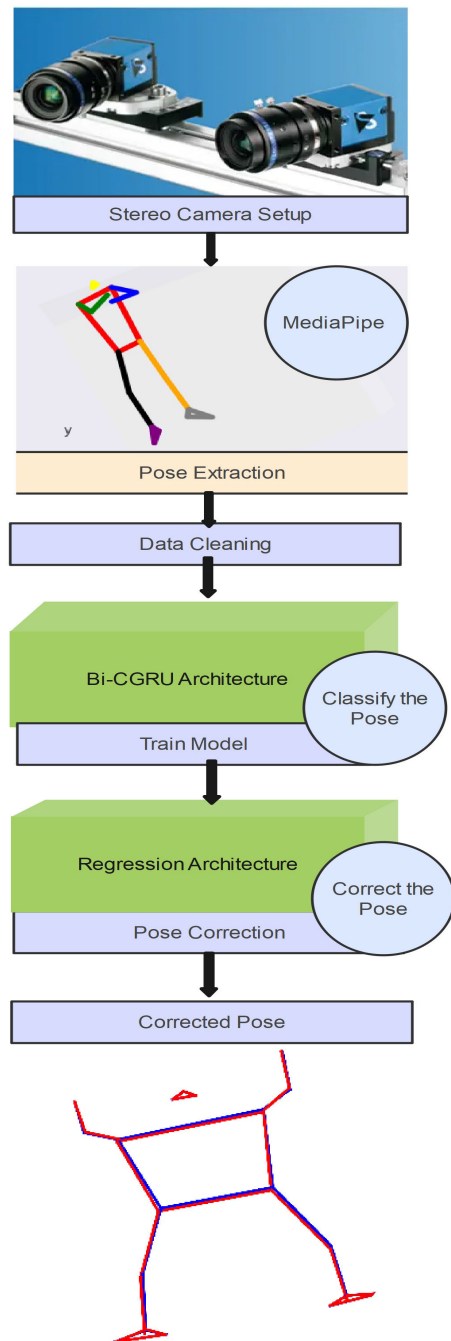


FIGURE 1. Block diagram of the proposed method.

NNs for squat analysis is investigated in [26]. Additionally, the authors in [27] integrate MediaPipe pose estimation with You Only Look Once (YOLO)'s pose estimation techniques to translate physical body movements into virtual environments. Similarly, [28] presents a system that utilizes MediaPipe and OpenCV for real-time posture correction, while [29] estimates the 3D pose and provides users with visual feedback to improve their performance. In the domain of yoga, AI algorithms have been developed to deliver feedback, including voice-activated reminders and personalized session engagement [30]. Furthermore, [31]

introduces Machine Learning (ML)-based methods for analyzing yoga poses. Advanced techniques leveraging MediaPipe to extract joint angles from webcam data and compare them to expert-defined yoga joint angles are proposed in [31].

To assess squat performance, the integration of Kinect camera-based systems with a Wii Balance Board has been investigated in [33]. A Naïve Bayes and SVM classifier is utilized to classify squat postures as either good or bad. Squat classification using SVM with MediaPipe pose estimation is explored in [34], while [35] proposes a Double Exponential Smoothing (DES) utilizing a single-camera with MediaPipe-extracted poses. The study in [36] investigates ML and pattern recognition techniques to identify full-body movement patterns in deep squats. Using HMM and CNN, [37] and [38] classify the squats into six distinct categories based on motion and visual data. In [11], a method employing a 1 Dimensional (1D) CNN is introduced to categorize squats into seven distinct types after normalizing pose data using node distances. The USquat system, presented in [2], applies computer vision and ML for squat analysis. Furthermore, [39] employs motion analysis with OpenPose for squat assessment. A squat motion detection model integrating MediaPipe with You Only Look Once v5 (YOLOv5) is proposed in [40], successfully detecting deep squats while reducing false detection rates.

In [41], a technique for assessing the quality is proposed by generating a fitness dataset and utilizing unlabeled video data. The introduction of UCF101 Human Actions dataset (UCF101) [42], which comprises 101 distinct activities, including bodyweight squats, has contributed to advancements in action recognition research. Various deep learning techniques have been developed for action classification, including LSTMs, a variant of Recurrent Neural Network (RNN) designed to retain short-term context [43], and Gated Recurrent Units (GRU), another variation of LSTM. For NN-based sequence translation, [44] introduces an attention mechanism known as Luong-style attention or dot-product attention. Alternative attention models, such as Bahdanau-style attention, have been explored in [45].

The proposed research evaluates expected limb lengths in squat performance to ensure a fair and standardized assessment for all individuals. The methodology integrates regression for squat correction, an encoder-based architecture for squat classification, and MediaPipe for pose estimation using squat data captured with a stereo camera. The key contributions of this research are as follows:

- 1) We utilize the MediaPipe algorithm to extract keypoints from videos recorded using a stereo camera.
- 2) We implement a custom Bi-CGRU deep learning encoder-decoder framework to evaluate squat performance and determine whether individuals perform squats correctly based on video data.
- 3) The research helps beginners assess their squat performance, identify squat types, and track their body posture during squat exercises.

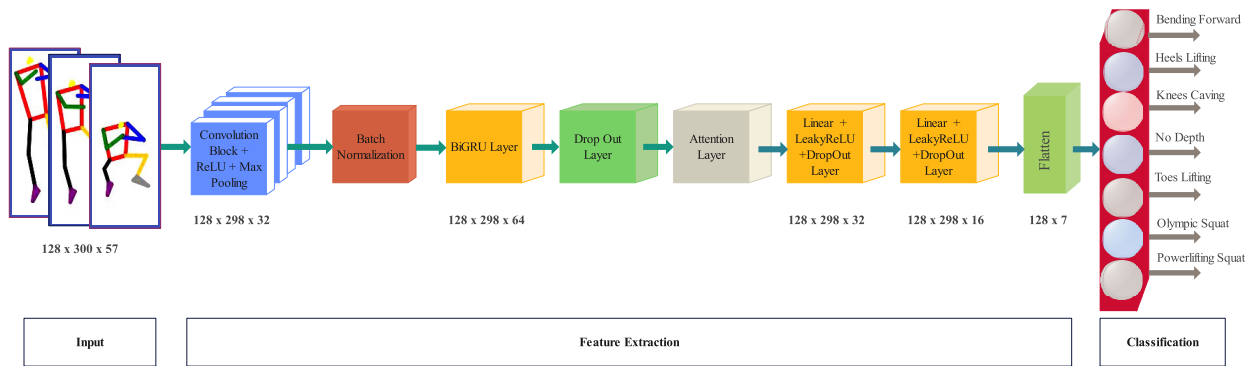


FIGURE 2. Convolutional Gated Recurrent Unit (Bi-CGRU) proposed architecture for squat classification.

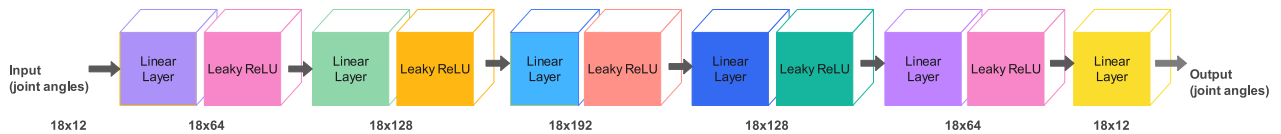


FIGURE 3. The proposed regression architecture for pose correction.

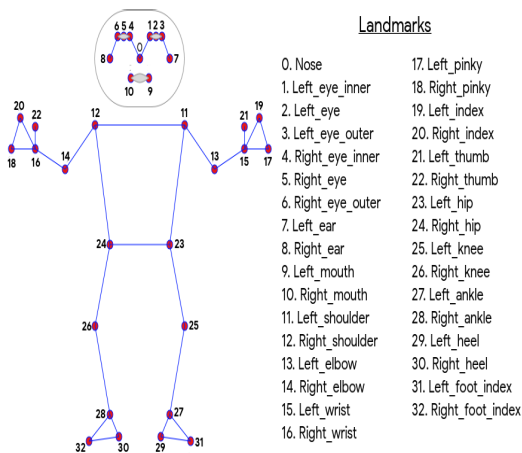


FIGURE 4. Squat position representation using skeletal model with 33 keypoints [32].

- 4) We develop a custom deep learning-based regressor to provide feedback, assisting individuals in understanding their technique and performing squats correctly.

Additionally, the study aims to propose a cost-effective system that detects the squat types and corrects improper squat execution by demonstrating the correct technique to users. This method reduces both time and cost, enabling individuals to perform squats effectively.

The manuscript is organized as follows: Section II outlines the proposed approach for classifying squat data types, error correction using a regressor, and provides a detailed explanation of each step, including the deep learning architecture and dataset collection procedure. Section III presents the analysis and results of the proposed approach,

comparing it with state-of-the-art methods. Finally, Section IV offers recommendations for future research in this area.

II. PROPOSED METHODOLOGY

The squat is one of the most challenging exercises, and the various myths surrounding it often discourage beginners. Many individuals perform squats independently due to a lack of nearby facilities or the high cost of coaching. In a critical exercise like squat, even minor errors can significantly alter one's technique. If performed incorrectly, this exercise can lead to injuries. Several existing research relies on smaller sample sizes and controlled laboratory settings. However, to develop reliable and widely applicable AI models, there is a need for more diverse, real-world data that accounts for variations in body types, exercise habits, and environmental factors. By leveraging advanced technologies such as computer vision and deep learning, the proposed approach aims to assist beginners in improving their squat technique.

To ensure cost-effectiveness, the proposed model seeks to introduce fitness and exercise regimens to beginners by utilizing existing software and equipment. Focusing specifically on squats, it aids individuals in training more effectively while minimizing the risk of injury. The application of this research also support physiotherapists and sports scientists in analyzing, diagnosing, and treating patients with greater expertise. The main steps of this work include:

- 1) Uses the dataset collected from stereo camera as described in [32] to feed MediaPipe algorithm to extract the poses.
- 2) A dataset consisting of 1332 squats from various individuals was fed into a custom Bi-CGRU classifier

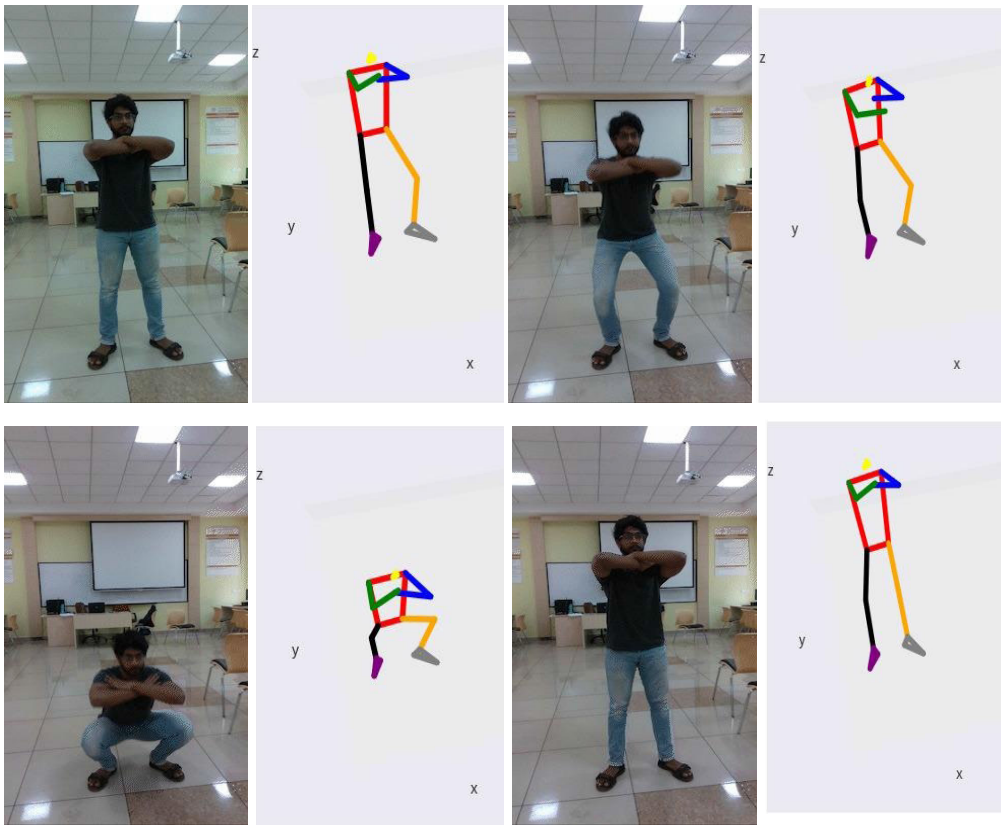


FIGURE 5. Output of Direct Linear Transform (DLT) algorithm on sample video.

to identify the type of squat, evaluate its accuracy, and assess whether the squat was performed correctly.

- 3) Identify improperly performed squats and apply a deep learning-based regression technique to generate the correct form. The user is then presented with a comparison between the corrected squat and the incorrectly performed squat.
- 4) Conduct a comparative analysis to assess the proposed method in relation to existing state-of-the-art approaches.

The proposed method for squat classification and error correction is illustrated in Figure 1. A stereo camera is used to capture depth information in the video. In contrast to prior research that uses a single camera with a side view, the current approach records video from a front view. The dataset is manually annotated to train the deep learning model. As shown in Algorithm 1, 3D video poses are extracted using MediaPipe. Data cleaning process is used to remove incomplete, irrelevant, duplicate videos. The dataset is split into training and testing subsets to train a custom autoencoder deep learning model. The robustness of LSTM has been demonstrated in various studies for predicting extreme floods [46] and river flow [47], as LSTMs excel at capturing long-term dependencies, albeit at the cost of higher complexity. LSTM is widely used in time series prediction

Algorithm 1 Squat Recognition and Correction Algorithm

1. Read the video from the stereo camera.
2. Feed the frames to the MediaPipe algorithm to extract the key points.
3. Input the pose key points and annotations into the Bi-CGRU network for training.
4. Train the regression network using the keypoints of correct squat posture.
5. Test the user data to check the accuracy and type of squat.
6. Show the corrected squat version to the user using pose diagram.

tasks. In the proposed research, we compare LSTM, RNN, and GRU for squat classification. Figure 2 illustrates the custom Bi-CGRU model proposed for classifying squat types. This approach enables accurate differentiation between various squat types. As shown in Figure 3, a deep learning-based regression technique is employed to correct incorrectly executed squats. To facilitate corrective training, participants are shown both the correct and incorrect squat forms. The code is implemented using TensorFlow and the OpenCV library in Python.

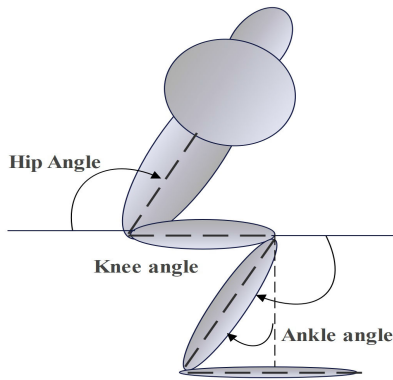


FIGURE 6. Squat with knee, hip, and ankle angles.

A. DATASETS

The dataset utilized in the study [32] is employed in the proposed approach. In [48], the authors have presented proper joint angles for horizontal squats. In this study, we have considered a dataset where each participant is expected to follow the proper knee (flexion), hip (flexion), and ankle (dorsiflexion) angles, which refer to proper squatting. The pose with these angles is depicted in Figure 6. Although squat form varies based on individuality, the permissible range of angles is acceptable. Specifically, horizontal squats have a knee joint flexion angle ranging from 113 ± 7 degrees, a hip joint flexion angle of 128 ± 9 degrees, and an ankle joint dorsiflexion angle of 23 ± 6 degrees. Our system uses a curated dataset of correctly performed squats, validated by fitness professionals. Individuals with varying body proportions performed these reference squats.

The pose data is generated from the recorded video [49]. We utilized a Realsense depth camera with a 30 fps frame rate, capturing depth information that includes the 3D spatial coordinates (X, Y, Z) of each tracked keypoint. The two cameras are separated with a horizontal distance of 40 cm and a height of 1.2m from the ground. The detailed explanation is presented in the paper [32]. This is essential for precise pose estimation, particularly under varying lighting conditions and partial occlusions. The implementation of the MediaPipe algorithm is expanded to extract keypoints such as the hip, knee, and ankle joints. Camera calibration is performed by scanning a checkerboard pattern. The DLT algorithm is used to convert pixel coordinates to 3D coordinates. Calibration is conducted separately for both cameras, and the process provides rotations, translations, distortion coefficients, Root Mean Square Error (RMSE), and the camera matrix. The distortion coefficients and camera matrices of both cameras are then used in stereo calibration. Pixel coordinates from both cameras are triangulated into 3D coordinates using several key points. The proposed method leverages the MediaPipe pose keypoint detector and two calibrated cameras to estimate the 3D body pose. The use of a stereo camera allows for capturing the front view image and obtaining accurate 3D pose data.

All squat videos captured with a stereo camera are processed through the MediaPipe algorithm, which extracts landmarks and converts them into posture data. Nineteen of the 33 key points identified by MediaPipe are extracted from the video data using this technique. These key points are illustrated in Figure 4. The intrinsic camera parameters for each camera are then computed, and the world coordinates are processed using a checkerboard pattern and the DLT algorithm. Unlike MediaPipe's normalized data, the pose data are adjusted to align with the subject in the video. The application of the DLT algorithm on skeleton and key point sample frames is shown in Figure 5.

Using a stereo camera, a diverse dataset consisting of 1332 videos, each lasting 4 to 5 seconds, was collected from fifty participants [32]. Consequently, over 1332 video snippets were gathered and classified into seven distinct categories: Olympic Squat (264), Knees Caving (152), Heels Lifting (177), No Depth (182), Toes Lifting (154), Bending Forward (163), and Powerlifting Squat (240). The length of the videos ranged from 30 to 291 frames. Based on this, it was determined that padding the data to 300 frames was optimal. Eighty percent of the dataset is allocated for model training, while the remaining twenty percent is reserved for validation.

B. SQUAT CLASSIFICATION MODEL

The proposed custom autoencoder model called Bi-CGRU architecture is depicted in the Figure 2 to classify squats into seven types.

1) CLASSIFIER MODEL

Time-series data is a critical component of squat analysis, as a user's posture dynamically changes throughout the squatting motion. When processing sequential data, Bi-CGRUs exhibit superior capability in capturing movement patterns and temporal dependencies. Although depth analysis is often represented as a single scalar value, it inherently requires an understanding of joint kinematics, particularly the coordinated motion of the hips, knees, and ankles. The convolutional layers of a Bi-CGRU facilitate the extraction of spatial features from input modalities such as motion capture data, enabling a more robust representation of movement dynamics. A bidirectional approach enhances performance by incorporating both past and future frames in the sequence, thereby allowing the model to detect subtle variations between the ascent and descent phases of the squat. This bidirectional architecture improves the precision in identifying key movement transitions, such as squat initiation and completion. Given the inherent variability in squatting mechanics—including depth, form, velocity, and execution style—Bi-CGRUs effectively model these variations by learning correlations in sequential data. Convolutional layers further refine feature extraction by filtering noise and emphasizing salient motion patterns. When trained on diverse datasets, Bi-CGRUs demonstrate strong generalization to inter-individual differences and hidden variations in squat execution. However, constructing an effective Bi-CGRU

model necessitates a large volume of labeled data, which can be both costly and time-intensive to acquire.

For classification, a stacked Bi-CGRU model, both with and without an attention mechanism, was selected. The primary motivation for utilizing Bi-CGRUs in time-series analysis is their ability to capture temporal dependencies and provide contextual information. By incorporating a bidirectional architecture, the model enables contextual flow in both forward and backward directions, enhancing its capacity to learn temporal correlations effectively. The proposed framework consists of a 'Luong' attention mechanism, three bidirectional GRU layers, and a 1-D CNN module for squat classification. The choice of GRU over other sequential models is justified by its computational efficiency and superior performance on shorter datasets. Given the inherently symmetric nature of the squat motion sequence, the bidirectional structure of the GRU is particularly advantageous, as it allows the model to leverage information from both past and future time steps, thereby improving the representation of squat motion patterns. The sequence model is preceded by a CNN block, which extracts spatial features before passing the processed representations to the stacked GRU layers. This hybrid architecture facilitates both spatial and temporal feature learning, contributing to improved classification accuracy.

A 1-D CNN and GRU block combination is employed to capture spatial features while preserving temporal dependencies within the sequence. Given that the task involves both spatial and temporal elements, an architecture capable of effectively modeling both dimensions is essential. The input to this model consists of the output from the DLT transformation and the pose estimation derived from MediaPipe. Specifically, the input data comprises the coordinates of key human joints for each frame of the video sequence, representing the subject's movement throughout the activity. This structured input enables the model to learn motion patterns and joint dynamics crucial for accurate classification.

$$c_t = CNN(x_t) \quad (1)$$

In Equation 1, x_t refers to the input data and c_t is the output after 1 D convolution. This output undergoes maximum pooling before it is fed to the Bidirectional Gated Recurrent Unit (Bi-GRU) stack as input.

$$y_t, h_t = GRU(c_t^*, h_{t-1}) \quad (2)$$

In Equation (2), c_t^* denotes the maximum pooling result applied to c_t , while h_{t-1} signifies the previous hidden state, initialized to zero at the start of training. The variables y_t and h_t represent the output and current hidden state of the GRU encoder unit, respectively. The form of y_t , implemented using PyTorch, is observed to be twice as large as the hidden layer output stated during model startup. This increase in size occurs because outputs from both forward and backward passes through the input sequence are concatenated. The spatiotemporal aspects of the input

data are efficiently captured by this integrated CNN and GRU architecture. The attention layer then processes these features to calculate feature weights, emphasizing the key characteristics that differentiate various squat types.

Weights generated by the attention layer emphasize significant features while filtering out less relevant ones. Due to its simplicity and computational efficiency, this approach selects the Luong-style multiplicative attention mechanism over other available attention mechanisms. It demonstrates superior performance compared to contemporary attention mechanisms while requiring relatively fewer parameters for training. The Luong attention mechanism employs three scoring functions in its design: (i) a dot product between the decoder hidden state and all encoder hidden states, (ii) the computation of the softmax score for these dot products, as shown in Equation (4), and (iii) the multiplication of the resulting scores by their corresponding encoder hidden states, as given in Equation (3). The decoder hidden state is then concatenated with the sum of the scaled encoder hidden states, forming an input to a neural network that subsequently processes the propagated features.

$$s_{ij} = f(h_i, d_j) \quad (3)$$

$$a_{ij} = \text{softmax}(s_{ij}) \quad (4)$$

In this context, let $f(x)$ denote the operation of the dot product between the hidden states of the encoder h_i and the hidden state of the decoder d_j . The resulting dot products are then converted to weights a_{ij} within the range of 0 to 1 through the softmax function.

$$c_j = \sum (a_{ij} * h_i) \quad (5)$$

$$o_j = \tanh(W_c(c_j, d_j)) \quad (6)$$

The encoder's hidden states are eventually scaled by their respective weights to calibrate the importance of each state. The resultant weighted hidden states are aggregated to form a context vector (c_j). This context vector is then combined with the decoder's hidden state and input into a neural network. The generated output from this attention layer is subsequently passed to a dense layer network that classifies various types of squats. The architecture was trained for 1000 epochs using a T4 GPU, with a seed of 3407 for reproducibility and a learning rate of 0.001. The optimal model state was obtained via checkpointing, achieving a categorical accuracy of 96.1% on the validation dataset.

The angles and limb lengths are calculated for these coordinates, and the loss is computed considering two factors: the difference between the initial limb lengths and the limb lengths determined using the adjusting coordinates data, and the difference between the computed angles and the model's adjusted angles output. The optimizer aligns the calculated angles with the modified angles and iteratively modifies the coordinates while maintaining constant limb lengths. The model employs an attention mechanism. By giving various places weights, it enables the model to concentrate on particular input components. The human tendency to focus

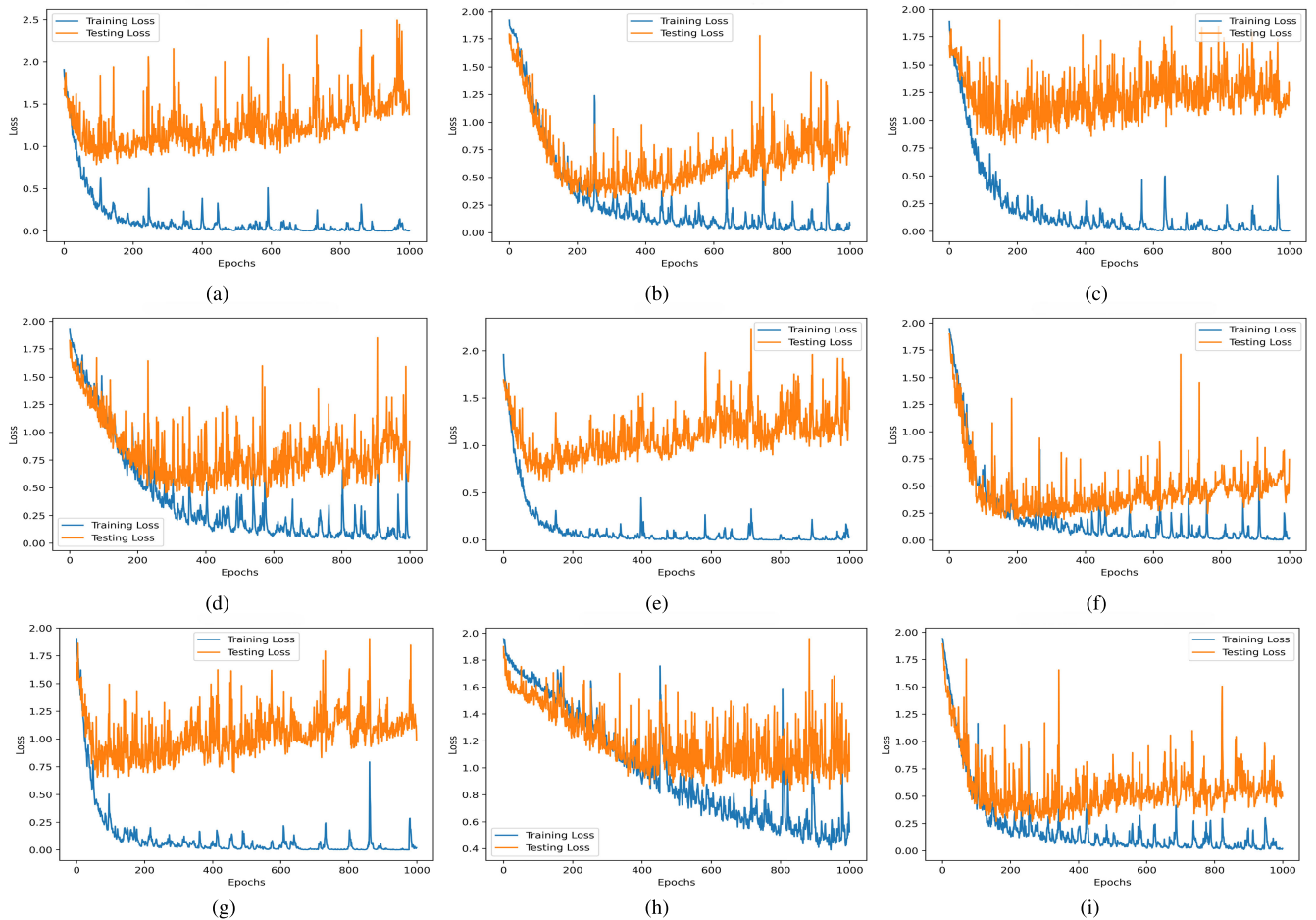


FIGURE 7. Loss versus Epoch of the models (a) Bi-RNN (b) Bi-LSTM (c) Bi-GRU (d) Bi-RNN with Attention (e) Bi-LSTM with Attention (f) Bi-GRU with Attention (g) Bi-GRU with Attention (h) Bi-GRU with Attention (i) Bi-RNN with Attention.

TABLE 1. Parameters of the proposed model.

Bi-CGRU Model	
Parameter	Value
Epochs	1000
Learning Rate	0.001
Batch Size	128
Optimizer	'adam'
Loss	categorical_crossentropy
Metrics	categorical_accuracy

on important details while disregarding less important ones is modeled by attention. This model makes use of Dot Product Attention or Luong type attention. The model details are given in Table 1.

2) REGRESSION ARCHITECTURE

The proposed regression network estimates joint angles and 2D or 3D coordinates for key anatomical joints, including the knees, hips, and shoulders. Contemporary methods predominantly focus on calculating joint angles and comparing them with an ideal reference posture to

correct deviations. A potential enhancement of this approach involves focusing on the angular relationships between body segments, which could offer a more compact and comprehensive representation of overall posture. In this framework, the model would assess posture holistically by comparing predicted and actual postural configurations. Future advancements may incorporate voice- or text-based corrective feedback, such as “maintain a straight back” or “increase the angle between knee joints,” enabling more effective user guidance. Furthermore, the integration of sensor-based feedback systems, which involve attaching sensors to specific joints, could substantially improve both the model’s accuracy and the user’s overall experience.

The proposed method classifies squats as either “good” or “bad.” The architecture utilizes MediaPipe-extracted landmarks to identify and categorize deviations from proper squat forms. To train the regressor, position data corresponding to correctly executed squats of a specific type is aggregated. Initially, the regressor model receives a list of joint angles representing a properly performed squat. A neural network-based regression model then fits curves to the data corresponding to various squat types, enabling the

TABLE 2. Quantitative analysis of the state-of-the-art techniques.

Architecture	Accuracy
Bi-RNN	77.52%
Bi-LSTM	85.76%
Bi-GRU [32]	83.52 %
RNN + Attention	82%
Bi-RNN + Attention	79.77%
Bi-LSTM + Attention	92.88%
Bi-GRU + Attention [32]	94 %
Bi-CLSTM	90.10%
Bi-CGRU + Attention	96.10 %

model to compare an individual's squat performance with the optimal execution predicted by the model. This process facilitates both assessment and correction of the squat form. A key application of this method is the powerlifting squat, where angles at key human joints are computed for each frame of video data collected during powerlifting squats. By connecting the points detected through MediaPipe pose estimation, line segments are formed, and the angles between these segments are calculated. Joint angles are preferred over raw coordinate data due to the complexities introduced by the variability in human body geometries. However, this approach may encounter challenges due to inaccuracies in joint angle estimation, which can be mitigated by applying a predefined threshold or tolerance. The neural network is trained on these joint angles, provided as sliding windows of frames, to predict subsequent windows and improve squat classification and correction.

The regression model captures the variation in joint angles across multiple instances of the same squat type over 200 epochs, with the best-performing model achieving a loss of less than 0.05 radians. For new squat samples, joint angles are computed and input into the trained model for inference, where the model generates corrected angles for each video frame. Using the Adam optimizer, an inverse operation is then applied to adjust the coordinate data according to the modified angles. The optimizer treats the initial coordinate data as adjustable parameters, iteratively refining the model's predictions to minimize discrepancies between the predicted and actual angles.

III. RESULTS AND DISCUSSION

For each joint (hip, knee, and ankle), we consider the 3D angular displacement vectors. Hip joint is the 3 DOF that includes flexion/extension angles that presents the primary movement in sagittal plane, hip abduction/adduction angle focusing the frontal plane, hip internal/external rotation angle. Similarly, the knee joint is the 2 DOF where primary knee flexion/extension angle and knee rotational angle are considered. The knee joint angle is related to length of the femur, length of tibia, and distance between hip and ankle centres. The ankle joint is also 2 DOF including ankle dorsiflexion/plantarflexion angle and inversion/eversion angle.

To develop the proposed approach, we utilized the dataset from [32]. The primary objective was to compare the performance of the models analyzed in [32] with our proposed model. All models were trained on the dataset without additional preprocessing and were replicated using PyTorch based on the model architecture described in [32]. Table 2 presents a performance comparison between the proposed model and those from [32].

The Bi-CGRU with an attention mechanism achieves the highest categorical accuracy on the test set, as shown in Table 2. Performance further improves when a CNN block is added before the sequence block. Figure 7 presents the loss curves for Bidirectional Long Short Term Memory (Bi-LSTM), Bi-GRU, Convolutional Long Short Term Memory (ConvLSTM), and Bidirectional Convolutional Gated Recurrent Unit (CGRU) models, both with and without the Luong attention layer. In addition to categorical accuracy and cross-entropy loss curves, the confusion matrix provides further insight into model performance. It evaluates the model's generalization ability and effectiveness in distinguishing between different classes. Figure 8 illustrates the confusion matrices for these models, categorizing predictions as true positives, true negatives, false positives, and false negatives by comparing them to actual labels.

The paper [32] discussed curve fit method to correct the squat posture. In this, the correct squat is considered as reference. The performed squat pose is fit into reference squat posture using curve fitting method. The curve connects keypoints from one frame to other frame as depicted in the Figure 9 (a-f). We have separated x, y, and z coordinates of Hip, Knee and Ankle joint keypoints. The blue color represents the tracked keypoints of a person performing the squat, while orange color represents the adjustment done to present the correct squat to the user. The displacement between actual and predicted is measured in pixels as depicted in the figures. These keypoints are combined to present the skeleton form to the user as predicted in the Figure 10. The detailed steps are as follows: (i) Select the proper squats. (ii) Extract the MediaPipe data (iii) Train the regressor to predict the joint angles and project the actual joint angles to the predicted angles. (iv) red color in Figure 9 shows actual squat and blue color shows corrected squat forms. This process is repeated for both the cameras to present 3D skeleton diagram. Three figures are presented to evaluate the regression model's performance in estimating squat postures. Figure 9 compares the predicted and actual joint angles for the hip, knee, and ankle joints of a subject performing a powerlifting squat. Based on these adjusted angles, Figure 10 visualizes the corrected coordinates of the subject's ankles, knees, hips, and shoulders. The training loss curve of the regression model on powerlifting squat data is shown in Figure 11, demonstrating moderate performance. While this research can be extended to other types of exercises, improvements in data ingestion, processing, and estimation are necessary to develop a more comprehensive end-user solution. Despite its efficiency and reliability,

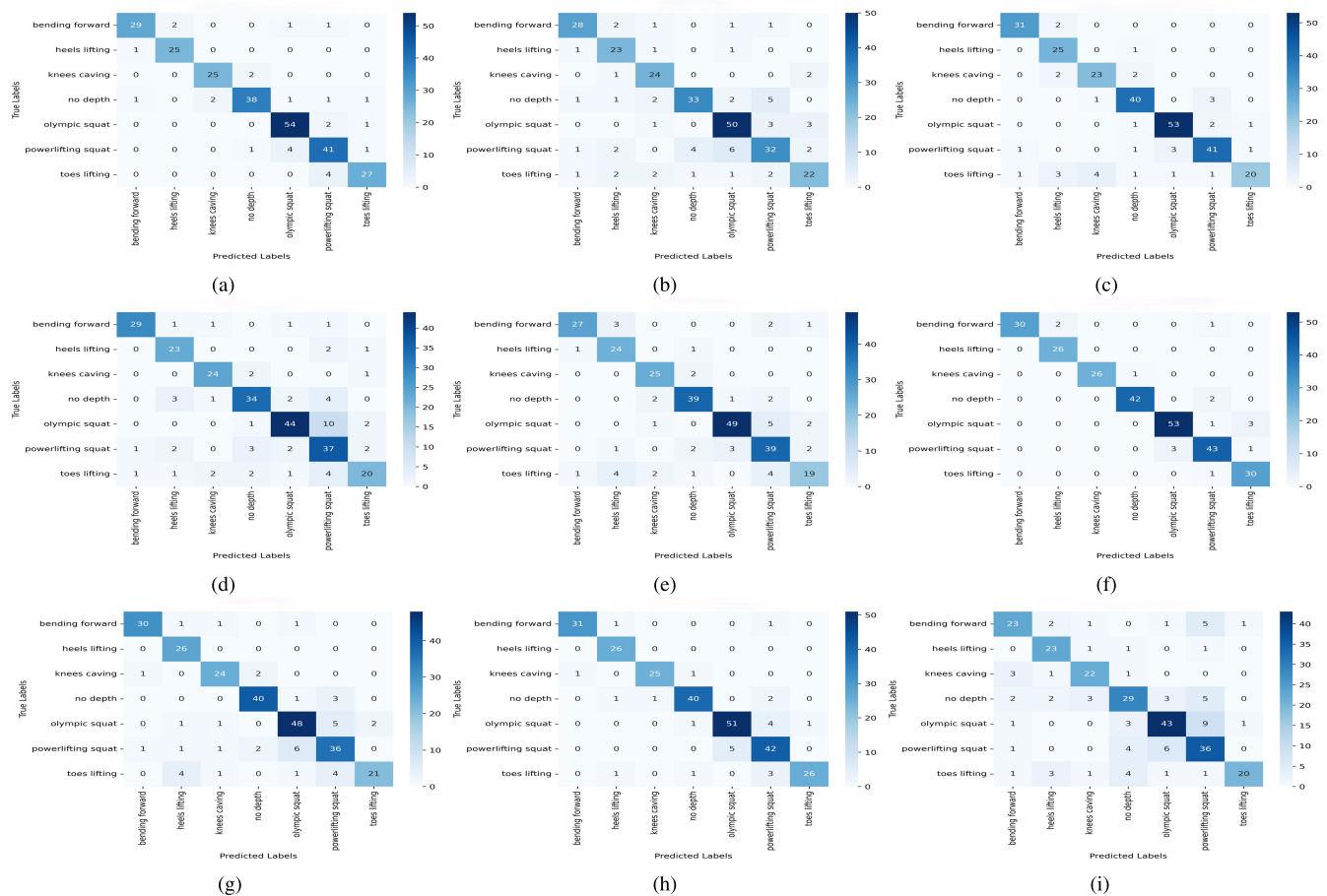


FIGURE 8. Loss versus Epoch of the models (a) Bi-GRU with Attention (b) Bi-GRU (c) Bi-LSTM with Attention (d) Bi-LSTM (e) CGRU (f) CGRU with Attention (g) ConvLSTM (h) CLSTM with Attention (i) Bi-RNN with Attention.

MediaPipe's pose estimation introduces significant noise when extracting postures from video frames. Preprocessing techniques, such as data cleaning and Gaussian smoothing, are essential for reducing these oscillations and enhancing the accuracy of subsequent analyses.

Since end users are more likely to utilize single-camera, developing a practical and reliable method to estimate depth (z-coordinate) using a monocular camera system instead of a stereo camera would be beneficial. Additionally, a faster and more lightweight approach could be explored for real-time estimation and correction, particularly for edge computing applications. While the neural network-based regression model for squat estimation and correction is expected to improve with a larger and more diverse training dataset that encompasses various types of squat, optimizing computational efficiency remains a key consideration.

Comparing the proposed method with existing squat analysis techniques is challenging due to variations in algorithms and datasets. To address this, we conducted a comparative study using advanced techniques on an available dataset, evaluating the strengths and limitations of each approach. Eight squat analysis algorithms were assessed from

both quantitative and qualitative perspectives. Our findings indicate that existing methods employ diverse datasets and approaches, often incorporating stereo cameras and deep learning techniques.

Evaluation of various models indicates that the Bi-CGRU model with an attention layer achieves the highest performance. Figure 12 presents the accuracy of Bidirectional Recurrent Neural Network (Bi-RNN), Bi-LSTM, and Bi-GRU models with and without attention layers. The Bi-CGRU model with attention demonstrates superior accuracy in both training and validation phases on the given dataset. Additionally, Figure 7 illustrates the loss curves for all models. Incorporating short-term memory context within long-term memory enhances model performance. Without an attention layer, the LSTM model outperforms the GRU model. However, with an attention layer, the GRU model surpasses the LSTM model. The RNN model exhibits lower accuracy, and its performance does not improve significantly with an attention mechanism. While models with attention layers perform well on Toes Lifting (TL), those without attention layers show suboptimal results. Among all models, the proposed Bi-CGRU achieves the highest accuracy of 96.1%, followed by the attention-enhanced

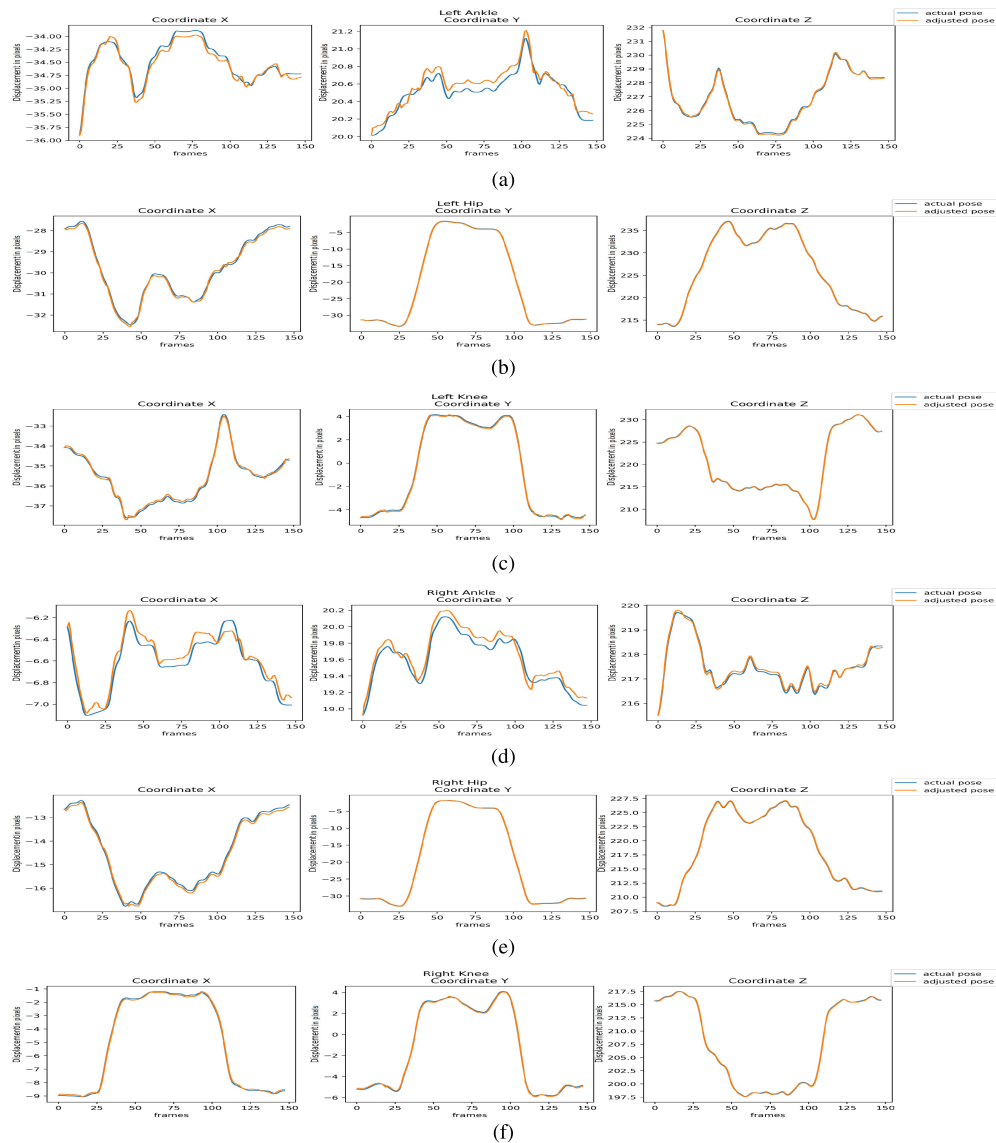


FIGURE 9. The predicted joint angles of the proposed regressor model.

Bi-GRU. In contrast, the Bi-RNN model records the lowest accuracy of 77.5% on the given dataset.

In Figure 10, the red skeleton represents an improper squat, while the blue skeleton illustrates the correct squat, enabling users to identify and correct their errors. Our correction model was trained on videos classified as good squat forms. To align the geometric points of an incorrect squat with those of a proper squat, the model employs a regression-based approach. This method provides users with the correct squat form, facilitating improved technique. Furthermore, we compare the proposed approach with [32], which utilizes a curve-fitting method to restore posture without preserving joint length aspect ratios. Unlike the curve-fitting approach, our method accurately represents squat posture. To the best of our knowledge, limited research has been conducted on squat analysis and correction using stereo cameras. While

previous approaches perform well in squat classification, they struggle to accurately represent proper posture. The curve-fitting method distorts proportional joint lengths when compared to the initial body position. In contrast, the proposed regression architecture, trained on various correct squat postures, effectively predicts joint angles for posture correction.

A. ABLATION STUDY

The proposed architecture consists of several key components, as illustrated in Figure 3. To assess the impact of the convolution block and attention unit, we conducted an ablation study by systematically adding and removing these components. Table 3 presents the model's accuracy with and without the convolution and attention blocks,

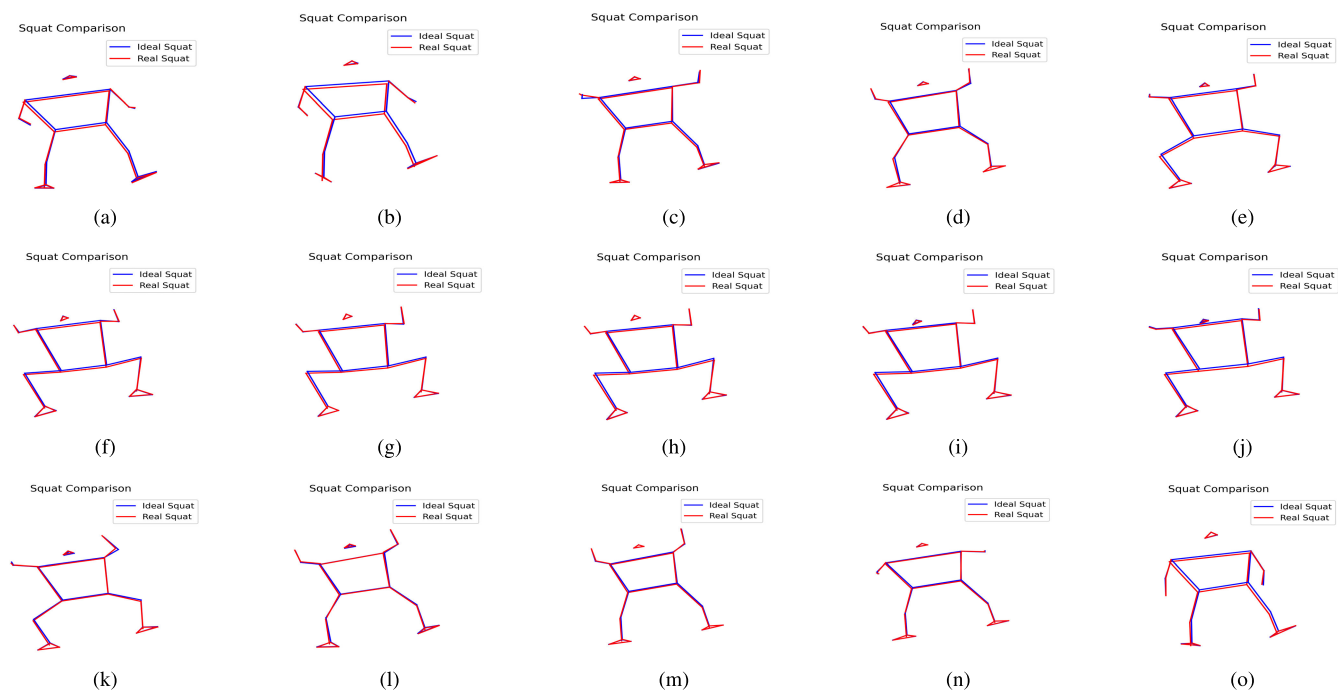


FIGURE 10. The skeleton showing the predicted squat performance by a right powerlifting squat type.

TABLE 3. Ablation study with Bi-GRU network.

Description	Bi-GRU Accuracy
Without Convolution Block and with Attention Block	94%
Without Convolution Block and Attention Block	83.52%
With Convolution Block and without Attention Block	88.80%
With Convolution Block and Attention Block	96.10%

demonstrating that their inclusion enhances the performance of the Bi-GRU model. The attention block, integrated into the encoder-decoder architecture, further improves the network’s performance. By leveraging hidden state information from each encoder, the attention mechanism acts as an interface between the encoder and decoder. This enables the model to learn relationships within the input sequence and selectively focus on relevant features, thereby enhancing accuracy and effectiveness.

B. LIMITATIONS AND FUTURE DIRECTION

Here’s a revised version with improved clarity, conciseness, and a more formal scientific tone:

Our model achieved an accuracy of 96.1%, surpassing the state-of-the-art approach reported in [32], which achieved 94%. This improvement is primarily attributed to the inclusion of a 1D convolutional block, which enhanced overall classification accuracy. Unlike previous studies, our approach also demonstrated greater robustness in presenting

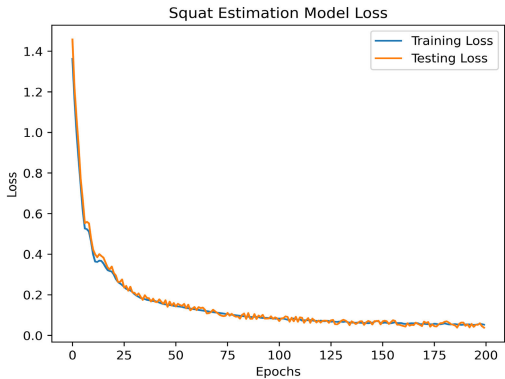


FIGURE 11. MSE loss curve of proposed regression architecture.

correct squat posture. The findings of this study have practical implications, particularly for mobile applications. For instance, if a mobile phone camera is used to capture squat videos, the model’s ability to correct posture can optimize squat form, leading to improved performance. This makes it a viable solution for professionals in the sports and fitness industry. However, challenges remain, such as the limitations of single-camera. We propose addressing this issue in future work by incorporating depth information alongside video data to enhance pose estimation accuracy.

The current study can be expanded by incorporating ML methods to analyze exercise movements, including applications in yoga and dance. In addition, the user could receive detailed explanations about workouts, highlighting

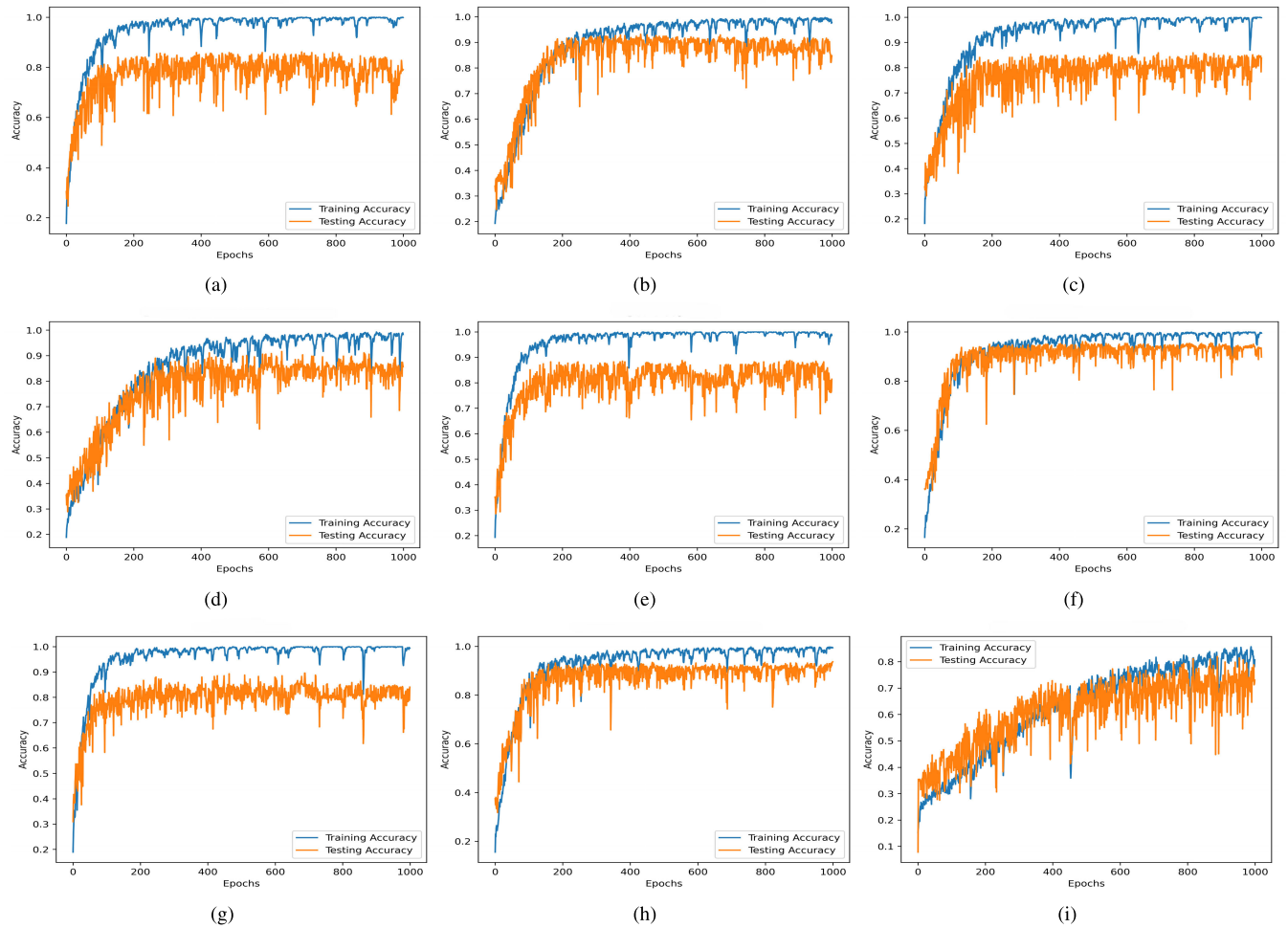


FIGURE 12. Accuracy vs epochs of the models (a) Bi-GRU (b) Bi-GRU with Attention Layer (c) Bi-LSTM (d) Bi-LSTM with Attention Layer (e) Bi-CGRU (f) Bi-CGRU with Attention Layer (g) Bi-CLSTM (h) Bi-CLSTM with Attention Layer (i) Bi-RNN with Attention Layer.

the benefits of proper form and the risks associated with improper execution. Nutritional guidance for optimal performance, such as diet recommendations for squats, could also be included as an added feature. The proposed approach utilizes regression analysis to adjust joint angles, presenting users with a skeletal model that helps with posture correction. To enhance user comprehension, a more intuitive representation, such as a manikin, could replace the stick-figure diagram. Furthermore, integrating exercise analysis into a virtual reality (VR) platform would increase engagement. Users could complete exercises and receive real-time corrections while wearing a VR headset, making the experience more immersive, interactive, and enjoyable.

The present study detects posture and records video using a stereo setup. This methodology could be expanded to utilize smartphone cameras for capturing an individual's posture. Furthermore, enhancing workout form could be facilitated by providing real-time, speech-based posture corrections to users. Ongoing research on squats and their variations is underway, and this work should be broadened to include other types of exercises. Several limitations in current sports

research can be addressed in future work: (i) Current AI algorithms typically provide generic feedback. Developing personalized models that account for individual biomechanics, fitness levels, and goals is crucial for tailoring workout recommendations and optimizing results. (ii) Defining and assessing exercise quality remains challenging. Objective measurements should consider factors such as range of motion, technique, and muscle activation, in addition to basic metrics like repetitions. (iii) Future research should aim to incorporate a diverse dataset representing various regions, human demographics, and workout types to improve the generalizability of the models.

IV. CONCLUSION

The proposed approach records user videos using a stereo setup and provides real-time posture corrections along with a pose diagram of the user's performance. We introduced a Bi-CGRU model that uses pose data extracted by MediaPipe from the video to classify squat forms. The Bi-CGRU model was compared against Bi-LSTM, Bi-GRU, and Bi-RNN models, both with and without an attention unit. To support

users in achieving their fitness goals without the need for personal trainers, we also proposed a deep learning-based regression model to predict correct posture from a squat video. The proposed method outperforms state-of-the-art techniques in the field, achieving an accuracy of 96.1%. This approach can be further expanded by increasing dataset diversity to include other physical exercises. Additionally, the method could be adapted for mobile devices, incorporating voice input for real-time workout analysis. As with any AI-based system, it is critical to ensure that training and evaluation data are of high quality and sufficiently diverse. Therefore, future training efforts must account for diversity in terms of geography, exercise types, and demographic groups. Environmental factors such as lighting and occlusion significantly affect the performance of the computer vision technique used. The current dataset includes squat videos from the Indian population, which may limit the generalizability of the results. While the proposed modeling approach is effective, further enhancements—such as incorporating speech-based, vision-based, or text-based inputs—could improve squat performance analysis. Future research should aim to expand the dataset to include a broader range of populations and exercises, including yoga and gym workouts, to better understand and test the method's adaptability across different contexts. Another promising direction is the integration of mobile phone cameras, which would enhance the real-time applicability of the approach.

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